# AI vs. Human in Refereeing Social Science

Pilot Summary: results from a trial run on 10 published papers

We asked an AI to review 10 papers that had already been through peer review at Nature Communications or Nature Human Behaviour. Because these journals publish the referee reports and the authors' replies, we can see what the human referees said and what the authors changed. We then compared the AI's review with the human reviews of the same paper. This summary explains how we did that and what we found.

## What we did

1. **Wrote the AI reviews.** For each paper, the AI read the preprint the authors posted before they submitted it to the journal, and wrote one referee report.
2. **Broke every report into critique points.** We split each report, human and AI, into critique points, one per concern. We then rated each point for how serious it is, how specific it is, whether it suggests a fix, and its tone.
3. **Built the issue record.** For each paper we listed the documented issues, meaning the problems that the review process took up. The list comes from the referee reports, the authors' replies, any editor letters, and what changed between the preprint and the published paper.
4. **Compared the reports with the issue record.** We checked which issues in the record each human referee and the AI raised. We also checked a sample of the AI's statements about each paper against the paper itself.

AI reports are much longer than human ones. To compare like with like, we show most results twice: once for the full AI report, and once for a length-matched AI report that keeps only as many critique points as the human referee wrote, taken from the top of the AI report.

## Terms used in this summary

- **Critique point**: one concern in one report about one thing in the paper, such as "the sample is too small" or "Table 2 has a typo".
- **Documented issue**: a problem with the paper that we can see was dealt with during review. It was raised by a referee or the editor, or the authors changed it between the preprint and the published version. If three referees raise the same problem, it is one issue.
- **Issue record**: the list of documented issues for one paper. We built it before comparing any AI report with it.
- **Unasked change**: something the authors changed between the preprint and the published paper although no referee or editor asked for it.
- **Emphasized issue**: an issue raised by two or more referees, asked for by the editor, or raised again by a referee in a later round.
- **Coverage**: the share of a paper's documented issues that a report raises. To be fair, we score each referee only on the issues someone else documented, so referees get no credit for issues that are on the list only because they raised them. The AI is scored on the same issues.
- **Same or related match**: an AI critique point is the "same" as a documented issue when it raises the same problem, so fixing one fixes the other. It is "related" when it is about the same part of the paper but makes a different point. Our headline numbers count only "same". The share of a report's critique points that are a "same" match is its alignment.

- **New issue**: when we add a second reviewer to one referee, a documented issue the first referee missed and the second reviewer raised. It counts once, however many critique points mention it.
- **Full and length-matched AI report**: the full AI report is the report as the AI wrote it. The length-matched AI report keeps only its first critique points, as many as the referee it is compared with wrote.
- **Factual error**: a statement in a report about what the paper says, such as a number, a quote or what a section contains, that the paper contradicts. We checked 5 to 10 such statements per AI report, 72 in all.

## Main findings

5. The AI writes a very different kind of report. It makes about 20 critique points per paper, against about 6 for a human referee. Its critique points are no more serious than a referee's, but they point to more specific places in the paper, almost always suggest a fix, and are always neutral in tone.
6. Taken as a whole, the AI report catches more than any one referee. It raised 27% of the documented issues, against 16% for a referee. Added to one referee's report, it brought in almost three times as many new issues as a second human referee did (2.8 against 1.0).
7. Point for point, the AI is about as good as a referee, not better. When we use the length-matched AI report, it raises 10% of the documented issues against the referee's 16%, and adds the same number of new issues as a second referee (1.0 against 1.0). Its advantage comes from writing more.
8. Referees are better at finding the problems that the review process cared most about. On emphasized issues, a referee raised 43%. The full AI report did about as well (44%), but the length-matched AI report raised only 11%. The AI puts its own priorities first, and they are often not the ones the referees agreed on.
9. The AI spots changes nobody asked for. It flagged a third of the changes authors made on their own between the preprint and the published paper. One example is a recalculated headline result in the forest-finance paper. No referee raised any of these changes.
10. Most of what the AI says is not in the review record, and what it says about the papers is accurate. 60% of its critique points match nothing that came up in review. We found no factual errors in the 72 statements we checked. Whether those extra critique points are useful is the main question for the full study.

## 1. How the reports differ

Figure 1 compares the average AI critique point with the average human one. The two are equally serious. The AI's critique points are more specific (2.94 against 2.54 on a scale of 1 to 3) and more often suggest a concrete fix (2.00 against 1.36 on a scale of 0 to 2). Human referees are a little warmer in tone; the AI is always neutral. Both spend about 70% of their critique points on the theory, data, methods and conclusions, and the rest on things like the paper's contribution, the literature and the writing.

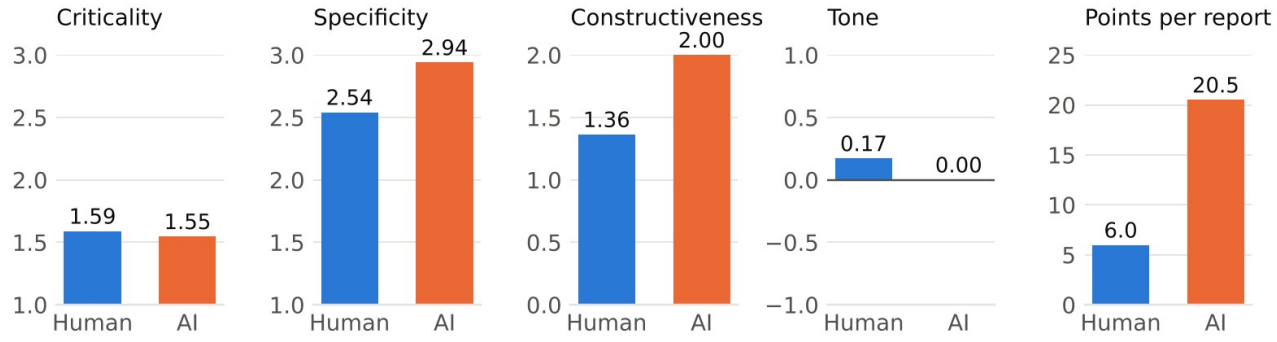



*Figure 1. Average ratings per critique point and number of critique points per report, human referees and AI (10 papers).*

## 2. Does the AI raise the problems that came up in review?

We matched each of the AI's 205 critique points to the list of documented issues (Figure 2). About one in six (17%) is a "same" match: it raises the same problem as a documented issue. Another 23% are about the same part of the paper but make a different point. The other 60% are about things nobody raised in review. Most of these ask the authors to check their results in other ways, point out numbers that don't agree across the paper, or ask for data and code.

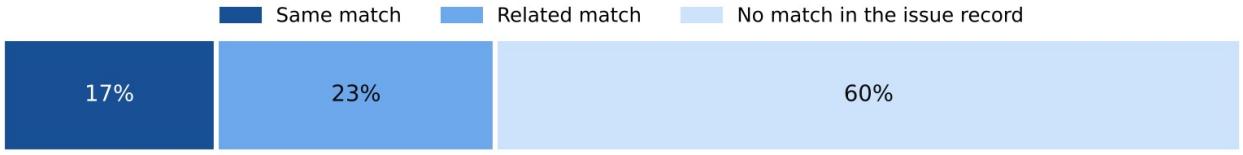



*Figure 2. How the AI's 205 critique points compare with the problems documented in review.*

Figure 3 shows coverage for different kinds of issues. The full AI report raises more documented issues than a referee in every group except emphasized issues, where the two are about even. The length-matched AI report falls behind the referee in every group except unasked changes, and it falls furthest behind on emphasized issues (11% against 43%).

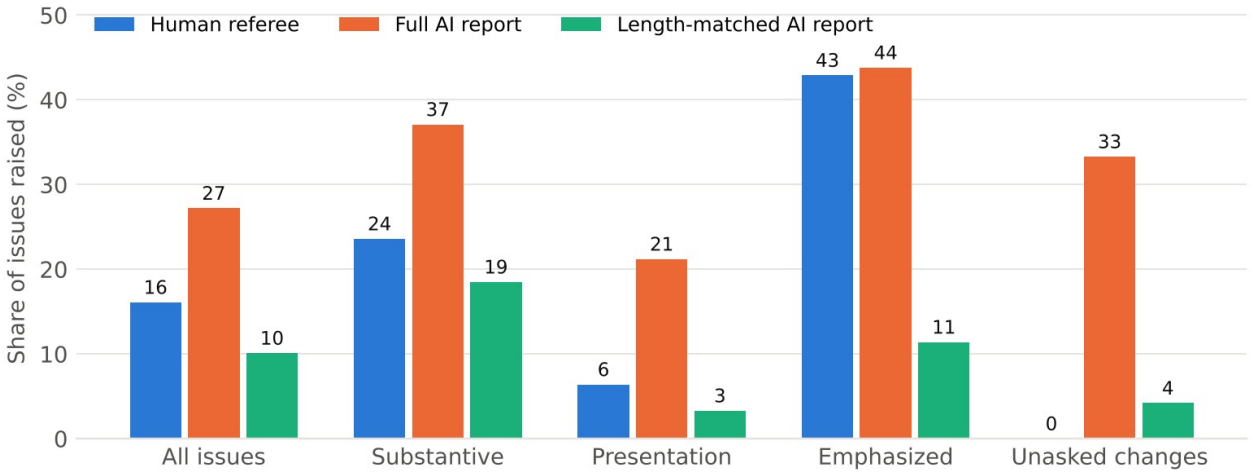



*Figure 3. Share of documented issues each report raises, by kind of issue (average over papers).*

Results vary a lot from paper to paper (Figure 4). The full AI report beats the average referee on 7 of the 10 papers. The length-matched AI report beats the referee on 5.

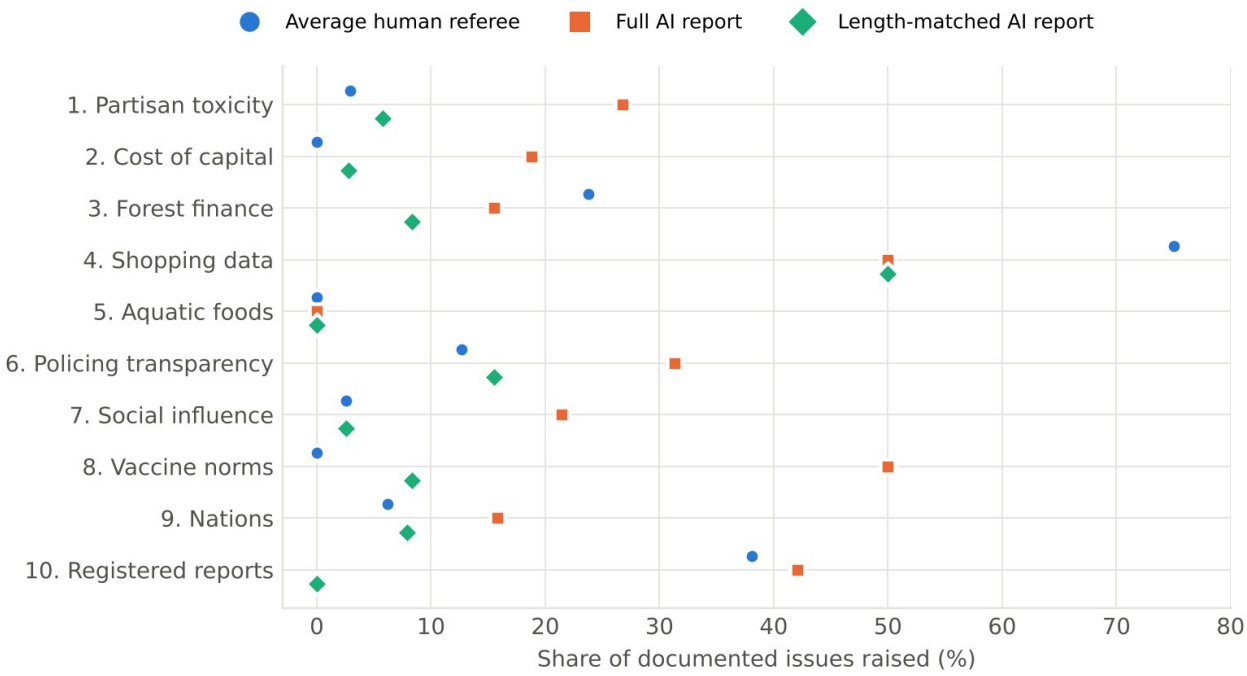



*Figure 4. Share of documented issues raised on each paper by the average human referee, the full AI report and the length-matched AI report.*

## 3. What does the AI add to a human referee?

Journals usually send a paper to two or more referees. So we asked what a second reviewer adds. For each pair of referees on a paper (72 pairs across 9 papers), we counted how many new issues the second referee raised that the first had missed, and did the same with the AI in place of the second referee. The full AI report adds 2.8 new issues, against 1.0 for a second human referee. The length-matched AI report adds the same as a second human (1.0).

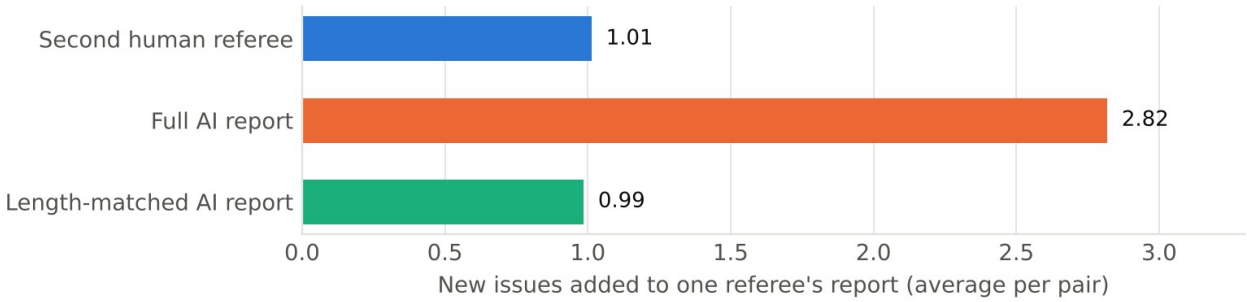



*Figure 5. New documented issues a second reviewer adds to one referee's report.*

## 4. How much do reviewers agree?

Two human referees on the same paper have only 7% of their critique points in common. The AI shares 8% of its critique points with a human referee. So referees rarely make the same points as each other, and the AI is no closer to any one referee than referees are to each other. To compare different AI reviews of the same paper, the full study will write several AI reviews per paper.

## 5. Accuracy, and where the AI does well

We checked 72 statements the AI made about the papers, such as numbers it quoted and claims about what a section says. We found no factual errors. In one paper, the AI also found two reporting errors that are still in the published article.

The table below shows coverage by type of study. On experiments, the full AI report raises almost three times as many documented issues as a referee (36% against 13%). On modelling and data papers, its lead is small (21% against 18%). With only 4 and 6 papers in each group, this shows the kind of comparison the full study will make rather than a firm result.

| Type of study | Papers | Human referee | Full AI report | Length-matched AI report |
|---|---|---|---|---|
| Experiments | 4 | 13% | 36% | 7% |
| Modelling and data papers | 6 | 18% | 21% | 13% |